Zhuochun Li

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Education

University of Pittsburgh — Pittsburgh, USA
Ph.D. in Information Science — Aug 2023 – Expected Summer 2027
Master of Science in Information Science, GPA: 3.9/4.0 — Jan 2021 – Dec 2022
School of Computing and Information (SCI)

Xi'an University of Technology — Xi'an, China
Bachelor of Engineering in Computer Science — Sep 2016 – Jul 2020

Profile

Highly motivated Ph.D. candidate in Information Science with research interests in **NLP** and **Machine Learning**. Proficient in Python, Machine Learning, and advanced NLP methods. Research focuses on **LLM Reasoning**, **Knowledge Distillation**, **Post-training, and LLM agents**. Completed **Applied Scientist Intern** at **Amazon** and **Microsoft**. Published research in top venues such as **SIGIR, EMNLP, ACL,** and **ICLR**.

Selected Publications [Google Scholar]

Li, Zhuochun, et al. "One Success Isn't Reliability: Thinkingbox, a Sandbox and Benchmark for Agents in Stateful Business Workflows." arXiv preprint arXiv:2608.19741 (2026). Submitted to ***ICLR 2027***.

Li, Zhuochun, Yuelyu Ji, Yiming Zeng, and Daqing He. "SPEAR: Distilling Domain-Adaptive Reasoning Skeletons via Sequential Symbolic Alignment in Reinforcement Learning." arXiv preprint arXiv:2608.26550 (2026). Submitted to ***NAACL 2027.***

Li, Zhuochun, et al. "Rethinking LLM-as-a-judge: representation-as-a-judge with small language models via semantic capacity asymmetry." ***ICLR 2026***.

César Guerra-Solano, **Zhuochun Li,** Xiang Lorraine Li. "Think Globally, Group Locally: Evaluating LLMs Using Multi-Lingual Word Grouping Games." ***EMNLP 2025***.

Li, Zhuochun, Yuelyu Ji, Rui Meng, Daqing He. "Learning from Committee: Reasoning Distillation from a Mixture of Teachers with Peer-Review." ***ACL 2025.***

Ji, Yuelyu, **Zhuochun Li**, Rui Meng, Daqing He. "Reason-to-Rank: Distilling Direct and Comparative Reasoning from Large Language Models for Document Reranking." ***SIGIR 2025***.

Ji, Yuelyu, **Zhuochun Li**, Rui Meng, Sonish Sivarajkumar, Yanshan Wang, Zeshui Yu, Hui Ji, Yushui Han, Hanyu Zeng, and Daqing He. "RAG-RLRC-LaySum at BioLaySumm: Integrating Retrieval-Augmented Generation and Readability Control for Layman Summarization of Biomedical Texts." ***ACL 2024.***

Li, Zhuochun, Khushboo Thaker, and Daqing He. "SiaKey: A Method for Improving Few-shot Learning with Clinical Domain Information." In *2023 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI)*, pp. 1-4. IEEE, 2023.

Li, Zhuochun, Wang, Zhixiao, Wenyao Yan, Min Huang, Qinyuan Fan, and Xin Wang. "Domestic Violence Crisis Recognition Method based on Bi-LSTM+ Attention." In *2022 8th Annual International Conference on Network and Information Systems for Computers (ICNISC)*, pp. 569-575. IEEE, 2022.

Academic Experience

PhD Dissertation, University of Pittsburgh — Oct 2026 – Present

Mutually Evolving Knowledge Distillation Framework for LLM Reasoning Capabilities. — *Pittsburgh, U.S.*

- Developed novel multi-teacher distillation and representation-based evaluation methods to improve reasoning supervision quality across mathematical, scientific, and commonsense reasoning tasks.
- Designed SPEAR, a lightweight symbolic process reward for on-policy reasoning distillation, leveraging domain-adaptive symbolic extraction to reward both outcome correctness and reasoning quality via RL.
- Proposed a **mutually evolving teacher–student distillation framework** integrating faithfulness verification, symbolic process alignment, and adaptive teacher guidance to improve reasoning distillation.

Research Assistant, iRiS Lab at University of Pittsburgh May 2022 – May 2023

Improve Few-shot Learning with Clinical Domain Information. *Pittsburgh, U.S.*

- Conducted the project of Ovarian Cancer Forum, which cooperated with the School of Nursing.
- Studied few-shot learning methods in the application of text classification and recommendation systems.
- Achieved average accuracy of over 60% by 10-shot training on Siamese Networks via triple loss function, which is comparable to the BERT performance using the whole dataset on this task.

Computer Vision - 16-720A, Carnegie Mellon University Jan 2022 - Apr 2022

Complete the Computer Vision course with grade A taught by Professor David Held of CMU *Pittsburgh, U.S.*

- Learned technologies about Spatial Pyramid Matching, Planar Homographies and Lucas-Kanade Tracking.
- Contributed to tasks about 3D Reconstruction, Neural Networks for Recognition and Photometric Stereo.
- Comprehended the cutting-edge deep learning models such as GAN, VAE and Transformers.

Undergraduate Thesis Jan 2020 - Jul 2020

Implementation of Domestic Violence Crisis Recognition Method Based on Deep Learning *Xi'an, China*

- Related paper has been published in the 2022 8th Annual International Conference on Network and Information Systems for Computers (ICNISC).
- Collected 1654 posts related to Domestic violence and built 50-dimensional word vectors via Word2Vec.
- Constructed CNN, RNN, LSTM, Bi-LSTM+self-Attention neural network models to accomplish text categorization task, Bi-LSTM+self-Attention model had the best performance, with an accuracy rate of 90.22% and recall rate of 93.98%.

Work Experience

Applied Scientist Intern, Microsoft May 2026 - Jul 2026

*Research on training and evaluation of **multi-turn long-horizon agents** in Copilot Studio* *Redmond, U.S.*

- Integrated the agent sandbox, **Thinkingbox**, with verl frameworks to improve **agentic RL**.
- Open-sourced **Thinkingbox-Bench**, a 507-task benchmark across five business domains, and established LLM performance and reliability leaderboards.

Applied Scientist Intern, Amazon Sep 2025 - Dec 2025

*Research with evaluation and **optimization of LLM agents*** *San Diego, U.S.*

- Improved natural language optimization of LLMs based on **GEPA** framework and AWS Bedrock platform.
- Implemented prompt learning rate and token usage controls to optimize training efficiency and cost.

Research Intern, Ping An Technology May 2025 - Aug 2025

Research Internship about LLM Probing on reference-free evaluation tasks *Shenzhen, China*

- Published **ICLR 2026**: "Rethinking LLM-as-a-Judge: Representation-as-a-Judge with Small Language Models via Semantic Capacity Asymmetry".
- Probed small LMs' layer representations evaluation scores across 5 aspects on {question, answer} pairs.
- Built effective probing classifiers to filter data and improve fine-tuning performance on LLMs.

Machine Learning Engineer Intern, MEDA AI May 2022 - Aug 2022

Internship with Startup Half Moon Tech to work on the Meda Metaverse project *Remote, U.S.*

- Improved the text-to-speech (TTS) based on the Tacotron model and built server to train over 10M steps.
- Assisted in constructing 3D character model and got average accuracy over 85% on facial attributes classification task on dataset CelebA.
- Developed API for integrating with backend services and maintained Linux servers.

Software Engineering Intern, Pactera Technology May 2021 - Aug 2021

Design and Development of an Intelligent Customer Service System *Wuhan, China*

- Conducted field surveys over 100 customers, assisted designing database E-R model containing 34 tables.

- Contributed more than 20 web page interface implementations for different service requirements of clients.
- Developed online semantic analysis system using Baidu voice recognition API with 80% code coverage.

Skills

Programming Language/Platform: Python (main), C, C++, JAVA, SQL, Linux, GIT, AWS, Bedrock, Azure
Tools: Pytorch, Transformers, DeepSpeed, FSDP, Trl, Verl, Huggingface, Numpy, Pandas, Scikit-learn, Spacy.

Teaching

Fall 2026: TA for CS 1675 Intro to Machine Learning
Spring 2026: TA for CS 1671 Human Language Technologies
Spring 2025: TA for INFSCI 2440 Artificial Intelligence
Fall 2024: TA for INFSCI 0201 Intermediate Programming Python (LAB LEADER)
Fall 2024: TA for INFSCI 0410 Human Centered Systems
Spring 2024: TA for INFSCI 2440 Artificial Intelligence
Fall 2023: TA for INFSCI 2140 Information Storage & Retrieval
Fall 2023: TA for INFSCI 2410 Intro to Neural Networks

Professional Service

Reviewer, ICLR 2027
Reviewer, ARR August 2026
Reviewer, NeurIPS 2026
Reviewer, ACL 2026
Reviewer, ASIS&T 2026
Reviewer, ICHI 2026

Awards

Catherine Ofiesh and Gerald Orner Award (2026) - PhD academic excellence and top-tier research publication
Outstanding Undergraduate Thesis Award (2020)
Third Prize XUT Scholarship (2018)